

Artificial Intelligence

Lecture 04 – Logical Agents

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Basic Actions

- A knowledge base keeps track of things
- We can TELL it facts or ASK for inference
- For e.g.,
 - TELL: Father of John is Bob
 - TELL: Jane is John's Sister
 - TELL: John's Father is the same as John's sister's father
 - ASK: Who's Jane's Father?

Knowledge Based Agents

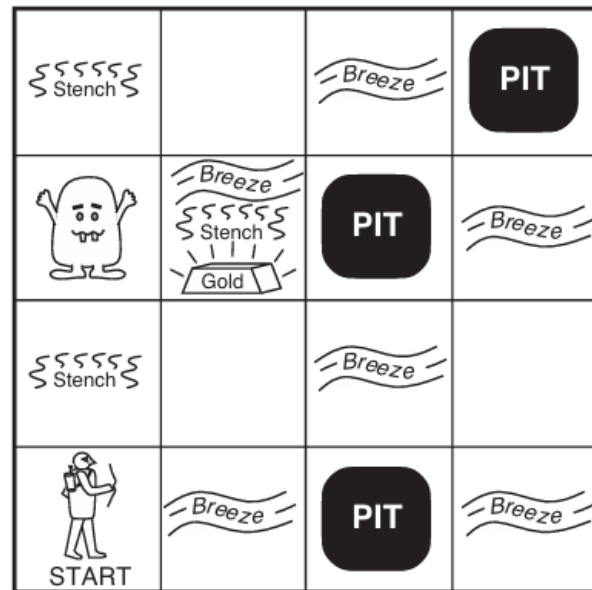
- Knowledge Base
- Knowledge Representation Language
- Inference
- Background knowledge about the world

Knowledge Based Agents

- At every step:
 - Construct a sentence with assertion about precepts
 - Construct a sentence asking what action is next
 - Constructs a sentence asserting that action

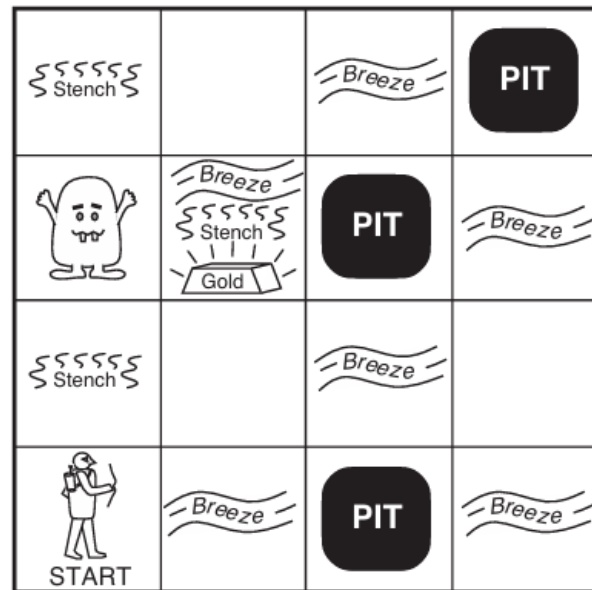
Wumpus World

- Adjacent rooms are connected (horizontally and vertically)
- Wumpus – a dangerous creature lurking in the cave
- Player can smell the “stench” of Wumpus
- Player feels a breeze if pit is nearby
- Player can shoot ONE arrow at the Wumpus to kill it
- Some rooms contain pits that will trap player
- One room contains a pot of gold



Wumpus World Formulation

- PEAS:
 - **Performance Measure:**
 - +1000 for walk-out with gold;
 - -1000 for dying;
 - -1 for each action,
 - -10 for arrow
 - **Environment:** a 4x4 grid. Agent starts at [1,1]; gold and pits randomly distributed, etc.
 - **Actuators:** Agent can move forward, left or right
 - **Sensors:** {*Smell, Breeze, Glitter, Bump, Scream*}



Wumpus World - Inference

1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2 OK	2,2	3,2	4,2
1,1 A OK	2,1 OK	3,1	4,1

A = Agent
B = Breeze
G = Glitter, Gold
OK = Safe square
P = Pit
S = Stench
V = Visited
W = Wumpus

Starting State

Precepts: {None, None, None, None, None}

Wumpus World - Inference

1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2 OK	2,2 P?	3,2	4,2
1,1 V OK	2,1 A B OK	3,1 P?	4,1

A = Agent
B = Breeze
G = Glitter, Gold
OK = Safe square
P = Pit
S = Stench
V = Visited
W = Wumpus

Next State

Precepts: {None, Breeze, None, None, None}

Wumpus World - Inference

1,4	2,4	3,4	4,4
1,3 W!	2,3	3,3	4,3
1,2 A S OK	2,2 OK	3,2	4,2
1,1 V OK	2,1 B V OK	3,1 P!	4,1

A = Agent
B = Breeze
G = Glitter, Gold
OK = Safe square
P = Pit
S = Stench
V = Visited
W = Wumpus

Next State

Precepts: {Stench, None, None, None, None}

Wumpus World - Inference

1,4	2,4 P?	3,4	4,4
1,3 W!	2,3 A S G B	3,3 P?	4,3
1,2 S V OK	2,2 V OK	3,2	4,2
1,1 V OK	2,1 B V OK	3,1 P!	4,1

A = Agent
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V = Visited
W = Wumpus

Next State

Precepts: {Stench, Breeze, Glitter, None, None}

Logic Basics

- A logic has **syntax**
- Semantics define the truth of a sentence
- Models describe possible worlds. $M(\alpha)$ is a set of all models of α
- If sentence α is true for model $\mathbf{m}$, α satisfies $\mathbf{m}$

Logic Entailment

- Entailment is when a sentence follows from another $\alpha \models \beta$
- $\alpha \models \beta$ iff in every model where α is true, β is also true
- $\alpha \models \beta$ iff $M(\alpha) \subseteq M(\beta)$
- Examples:
 - $(x=0) \models (xy=0)$
 - $(p=\text{True}) \models (p \vee q)$
 - $\text{False} \models \text{True}$
 - $(p \wedge q) \models (p \vee q)$
 - $((p \leftrightarrow q) \vee r) \models (p \rightarrow q)$
 - $((q \rightarrow p) \models (p \leftrightarrow q) \vee r))$

Models and KBs

- Model the presence of pits in squares [1,2], [2,2] and [3,1] with the restriction α_2 of no pits in [2,2].
- Then does KB $\models \alpha_2$?
- In every model in which KB is true, α_2 is not necessarily true.

A

= Agent

B

= Breeze

G

= Glitter, Gold

OK

= Safe square

P

= Pit

S

= Stench

V

= Visited

W

= Wumpus

1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2	2,2 P?	3,2	4,2
OK			
1,1	2,1 A B OK	3,1 P?	4,1

Figure 7.3(b):

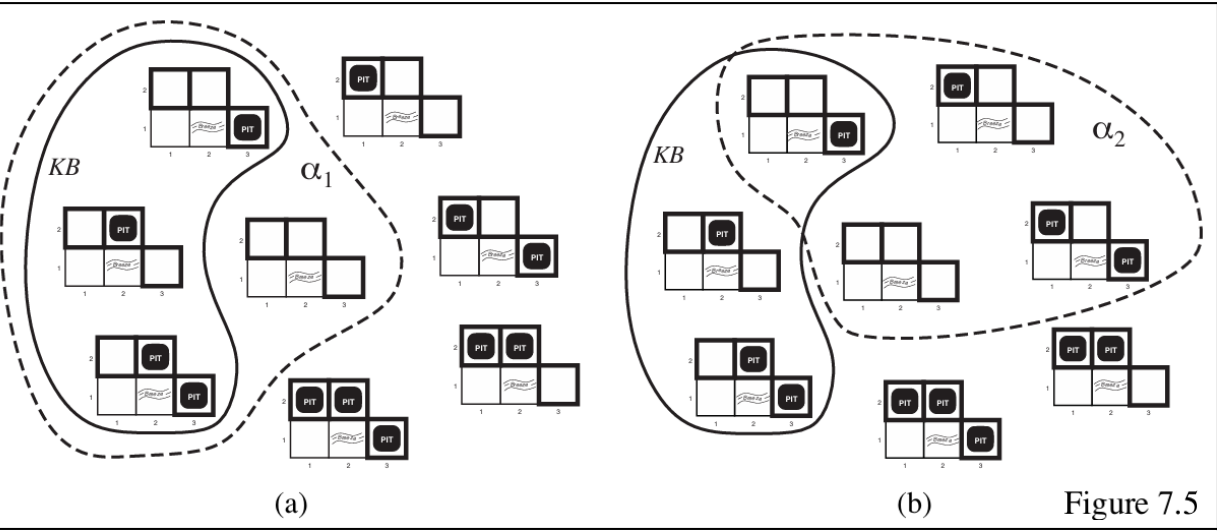


Figure 7.5

Semantics

- Sentences have a “truth value” w.r.t a model
- Example: $m_1 = \{ P_{1,2} = False, P_{2,2} = False, P_{2,1} = True \}$
- $P_{1,2}$ is just a symbol, it can mean anything
- Truth values of complex sentences can be evaluated recursively according to following 5 rules:
 - $\neg P$ is true iff P is false in m .
 - $P \wedge Q$ is true iff both P and Q are true in m .
 - $P \vee Q$ is true iff either P or Q is true in m .
 - $P \Rightarrow Q$ is true unless P is true and Q is false in m .
 - $P \Leftrightarrow Q$ is true iff P and Q are both true or both false in m .

Semantics – Truth Tables

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \Rightarrow Q$	$P \Leftrightarrow Q$
<i>false</i>	<i>false</i>	<i>true</i>	<i>false</i>	<i>false</i>	<i>true</i>	<i>true</i>
<i>false</i>	<i>true</i>	<i>true</i>	<i>false</i>	<i>true</i>	<i>true</i>	<i>false</i>
<i>true</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>true</i>	<i>false</i>	<i>false</i>
<i>true</i>	<i>true</i>	<i>false</i>	<i>true</i>	<i>true</i>	<i>true</i>	<i>true</i>

Semantics – Evaluating

In Model, $m_1 = \{ P_{1,2} = \textit{False}, P_{2,2} = \textit{False}, P_{2,1} = \textit{True} \}$

Evaluate Sentence: $\sim P_{1,2} \wedge (P_{2,2} \vee P_{2,1})$

Evaluate in Model, $m_2 = \{ P_{1,2} = \textit{True}, P_{2,2} = \textit{True}, P_{2,1} = \textit{False} \}$

A simple KB – Wumpus World

$P_{x,y}$ is true if there is a pit in $[x, y]$.

$W_{x,y}$ is true if there is a wumpus in $[x, y]$, dead or alive.

$B_{x,y}$ is true if the agent perceives a breeze in $[x, y]$.

$S_{x,y}$ is true if the agent perceives a stench in $[x, y]$.

Inference – Simple Method

enumerate the models, for each model check that:

If it is true in α it must be true in KB

- Models are assignments of true or false to every proposition symbol.

Rules – A simple KB

Wumpus-world relevant proposition symbols:

$B_{1,1}$, $B_{2,1}$, $P_{1,1}$, $P_{1,2}$, $P_{2,1}$, $P_{2,2}$, and $P_{3,1}$

For the Wumpus world in general,

(1) There is no pit in [1,1] $R_1 : \neg P_{1,1}$.

(2) A square is breezy if and only if there is a pit in a neighbouring square.

$R_2 : B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$.

$R_3 : B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$.

(3) True for all models

$R_4 : \neg B_{1,1}$.

$R_5 : B_{2,1}$.

A = Agent
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1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2 OK	2,2 P?	3,2	4,2
1,1 V OK	2,1 A B OK	3,1 P?	4,1

(b)

Inference - Goal

- ASK: Is there no pit in $[1,2]$?
 - i.e., does $KB \models \neg P_{1,2}$?
- We say that $\neg P_{1,2}$ is a sentence α
- Main goal: decide whether $KB \models \alpha$
- α can be much more complex query

enumerate the models, for each model check that:

If it is true in α it must be true in KB

$B_{1,1}$	$B_{2,1}$	$P_{1,1}$	$P_{1,2}$	$P_{2,1}$	$P_{2,2}$	$P_{3,1}$	R_1	R_2	R_3	R_4	R_5	KB
false	false	false	false	false	false	false	true	true	true	true	false	false
false	false	false	false	false	false	true	true	true	false	true	false	false
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
false	true	false	false	false	false	false	true	true	false	true	true	false
false	true	false	false	false	false	true	true	true	true	true	true	<u>true</u>
false	true	false	false	false	true	false	true	true	true	true	true	<u>true</u>
false	true	false	false	false	true	true	true	true	true	true	true	<u>true</u>
false	true	false	false	true	false	false	true	false	false	true	true	false
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
true	true	true	true	true	true	true	false	true	true	false	true	false

Inference – TT-Entails

```
function TT-ENTAILS?( $KB, \alpha$ ) returns true or false  
  inputs:  $KB$ , the knowledge base, a sentence in propositional logic  
            $\alpha$ , the query, a sentence in propositional logic  
  
   $symbols \leftarrow$  a list of the proposition symbols in  $KB$  and  $\alpha$   
  return TT-CHECK-ALL( $KB, \alpha, symbols, \{ \}$ )
```

```
function TT-CHECK-ALL( $KB, \alpha, symbols, model$ ) returns true or false  
  if EMPTY?( $symbols$ ) then  
    if PL-TRUE?( $KB, model$ ) then return PL-TRUE?( $\alpha, model$ )  
    else return true // when KB is false, always return true  
  else do  
     $P \leftarrow$  FIRST( $symbols$ )  
     $rest \leftarrow$  REST( $symbols$ )  
    return (TT-CHECK-ALL( $KB, \alpha, rest, model \cup \{P = true\}$ )  
            and  
            TT-CHECK-ALL( $KB, \alpha, rest, model \cup \{P = false\}$ ))
```

Inference – Model Checking Complexity

- If KB and α contain n symbols in all:
 - Time: $O(2^n)$
 - Space: $O(n)$ (depth-first)
- Propositional entailment is co-NP complete (probably no easier than NP-complete)

Propositional Theorem Proving

Logical Equivalence: $\alpha \equiv \beta$ if and only if $\alpha \models \beta$ and $\beta \models \alpha$.

$(\alpha \wedge \beta) \equiv (\beta \wedge \alpha)$	commutativity of $\wedge$
$(\alpha \vee \beta) \equiv (\beta \vee \alpha)$	commutativity of $\vee$
$((\alpha \wedge \beta) \wedge \gamma) \equiv (\alpha \wedge (\beta \wedge \gamma))$	associativity of $\wedge$
$((\alpha \vee \beta) \vee \gamma) \equiv (\alpha \vee (\beta \vee \gamma))$	associativity of $\vee$
$\neg(\neg\alpha) \equiv \alpha$	double-negation elimination
$(\alpha \Rightarrow \beta) \equiv (\neg\beta \Rightarrow \neg\alpha)$	contraposition
$(\alpha \Rightarrow \beta) \equiv (\neg\alpha \vee \beta)$	implication elimination
$(\alpha \Leftrightarrow \beta) \equiv ((\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha))$	biconditional elimination
$\neg(\alpha \wedge \beta) \equiv (\neg\alpha \vee \neg\beta)$	De Morgan
$\neg(\alpha \vee \beta) \equiv (\neg\alpha \wedge \neg\beta)$	De Morgan
$(\alpha \wedge (\beta \vee \gamma)) \equiv ((\alpha \wedge \beta) \vee (\alpha \wedge \gamma))$	distributivity of $\wedge$ over $\vee$
$(\alpha \vee (\beta \wedge \gamma)) \equiv ((\alpha \vee \beta) \wedge (\alpha \vee \gamma))$	distributivity of $\vee$ over $\wedge$

Figure 7.11 Standard logical equivalences. The symbols α , β , and γ stand for arbitrary sentences of propositional logic.

Propositional Theorem Proving

Validity: Deduction Theorem

For any sentences α and β , $\alpha \models \beta$ if and only if the sentence $(\alpha \Rightarrow \beta)$ is valid.

Satisfiability: A sentence is true in some model

Proof by contradiction

$\alpha \models \beta$ if and only if the sentence $(\alpha \wedge \neg\beta)$ is unsatisfiable.

• ...

Inference and Proofs

- Modus Ponens
$$\frac{\alpha \Rightarrow \beta, \quad \alpha}{\beta}$$

- AND-Elimination
$$\frac{\alpha \wedge \beta}{\alpha}$$

- Other Rules:

$$\frac{\alpha \Leftrightarrow \beta}{(\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)}$$

and

$$\frac{(\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)}{\alpha \Leftrightarrow \beta}.$$

Inference in Wumpus World

For the Wumpus world in general,

(1) There is no pit in [1,1] $R_1 : \neg P_{1,1}$.

(2) A square is breezy if and only if there is a pit in a neighbouring square.

$$R_2 : B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1}).$$

$$R_3 : B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1}).$$

(3) True for all models

$$R_4 : \neg B_{1,1}.$$

$$R_5 : B_{2,1}.$$

A = Agent
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1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2	2,2 P?	3,2	4,2
OK			
1,1	2,1 A B OK	3,1 P?	4,1
V OK			

(b)

Inference in Wumpus World

We have $KB = R_1 \wedge R_2 \wedge R_3 \wedge R_4 \wedge R_5$

We want to prove $\neg P_{1,2}$

$$R_6 : (B_{1,1} \Rightarrow (P_{1,2} \vee P_{2,1})) \wedge ((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1})$$

$$R_7 : ((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1}) .$$

$$R_8 : (\neg B_{1,1} \Rightarrow \neg(P_{1,2} \vee P_{2,1})) .$$

$$R_9 : \neg(P_{1,2} \vee P_{2,1}) .$$

$$R_{10} : \neg P_{1,2} \wedge \neg P_{2,1} .$$

A = Agent
B = Breeze
G = Glitter, Gold
OK = Safe square
P = Pit
S = Stench
V = Visited
W = Wumpus

1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2 OK	2,2 P?	3,2	4,2
1,1 V OK	2,1 A B OK	3,1 P?	4,1

(b)

Inference as Search

- INITIAL STATE: the initial knowledge base.
- ACTIONS: the set of actions consists of all the inference rules applied to all the sentences that match the top half of the inference rule.
- RESULT: the result of an action is to add the sentence in the bottom half of the inference rule.
- GOAL: the goal is a state that contains the sentence we are trying to prove.

Resolution

Lets say agent returns to [1,1] from [2,1] and goes to [1,2]

1,4	2,4	3,4	4,4
1,3 W!	2,3	3,3	4,3
1,2 A S OK	2,2 OK	3,2	4,2
1,1 V OK	2,1 B V OK	3,1 P!	4,1

We add:

$$R_{11} : \neg B_{1,2} .$$

$$R_{12} : B_{1,2} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{1,3})$$

We can continue to use the same process

$$R_{13} : \neg P_{2,2} . \qquad R_{15} : P_{1,1} \vee P_{2,2} \vee P_{3,1} .$$

$$R_{14} : \neg P_{1,3} . \qquad R_{16} : P_{1,1} \vee P_{3,1} .$$

Unit Resolution:

$$\frac{\ell_1 \vee \dots \vee \ell_k, \quad m}{\ell_1 \vee \dots \vee \ell_{i-1} \vee \ell_{i+1} \vee \dots \vee \ell_k} ,$$

where each ℓ is a literal and ℓ_i and m are **complementary literals**

Resolution Rule

The unit resolution rule can be generalized to the full **resolution** rule,

$$\frac{\ell_1 \vee \cdots \vee \ell_k, \quad m_1 \vee \cdots \vee m_n}{\ell_1 \vee \cdots \vee \ell_{i-1} \vee \ell_{i+1} \vee \cdots \vee \ell_k \vee m_1 \vee \cdots \vee m_{j-1} \vee m_{j+1} \vee \cdots \vee m_n},$$

Example:

$$\frac{P_{1,1} \vee P_{3,1}, \quad \neg P_{1,1} \vee \neg P_{2,2}}{P_{3,1} \vee \neg P_{2,2}}$$

Conjunctive Normal Form (CNF)

- Every sentence in propositional logic can be expressed as conjunction of disjunctions of literals.
- Example: convert into CNF: $B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$
 1. Eliminate $\Leftrightarrow$, replacing $\alpha \Leftrightarrow \beta$ with $(\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)$.
$$(B_{1,1} \Rightarrow (P_{1,2} \vee P_{2,1})) \wedge ((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1}) .$$
 2. Eliminate $\Rightarrow$, replacing $\alpha \Rightarrow \beta$ with $\neg\alpha \vee \beta$:
$$(\neg B_{1,1} \vee P_{1,2} \vee P_{2,1}) \wedge (\neg(P_{1,2} \vee P_{2,1}) \vee B_{1,1}) .$$

Conjunctive Normal Form (CNF)

3. CNF requires $\neg$ to appear only in literals, so we “move $\neg$ inwards” by repeated application of the following equivalences from Figure 7.11:

$$\neg(\neg\alpha) \equiv \alpha \quad (\text{double-negation elimination})$$

$$\neg(\alpha \wedge \beta) \equiv (\neg\alpha \vee \neg\beta) \quad (\text{De Morgan})$$

$$\neg(\alpha \vee \beta) \equiv (\neg\alpha \wedge \neg\beta) \quad (\text{De Morgan})$$

In the example, we require just one application of the last rule:

$$(\neg B_{1,1} \vee P_{1,2} \vee P_{2,1}) \wedge ((\neg P_{1,2} \wedge \neg P_{2,1}) \vee B_{1,1}) .$$

4. Now we have a sentence containing nested $\wedge$ and $\vee$ operators applied to literals. We apply the distributivity law from Figure 7.11, distributing $\vee$ over $\wedge$ wherever possible.

$$(\neg B_{1,1} \vee P_{1,2} \vee P_{2,1}) \wedge (\neg P_{1,2} \vee B_{1,1}) \wedge (\neg P_{2,1} \vee B_{1,1}) .$$

Resolution Algorithm

- Algorithm works using proof by contradiction
- To show that $KB \models \alpha$, we show that $(KB \wedge \neg\alpha)$ is unsatisfiable
- Apply resolution to $(KB \wedge \neg\alpha)$ in CNF
 - and resolve pairs with complementary literals

$$\frac{\ell_1 \vee \cdots \vee \ell_k, \quad m_1 \vee \cdots \vee m_n}{\ell_1 \vee \cdots \vee \ell_{i-1} \vee \ell_{i+1} \vee \cdots \vee \ell_k \vee m_1 \vee \cdots \vee m_{j-1} \vee m_{j+1} \vee \cdots \vee m_n},$$

- and add new clauses until
 - There are no new clauses to be added
 - Two clauses resolve to the empty class, which means $KB \models \alpha$

Resolution Algorithm

function PL-RESOLUTION(KB, α) **returns** *true* or *false*

inputs: KB , the knowledge base, a sentence in propositional logic
 α , the query, a sentence in propositional logic

$clauses \leftarrow$ the set of clauses in the CNF representation of $KB \wedge \neg\alpha$

$new \leftarrow \{ \}$

loop do

for each pair of clauses C_i, C_j **in** $clauses$ **do**

$resolvents \leftarrow$ PL-RESOLVE(C_i, C_j)

if $resolvents$ contains the empty clause **then return** *true*

$new \leftarrow new \cup resolvents$

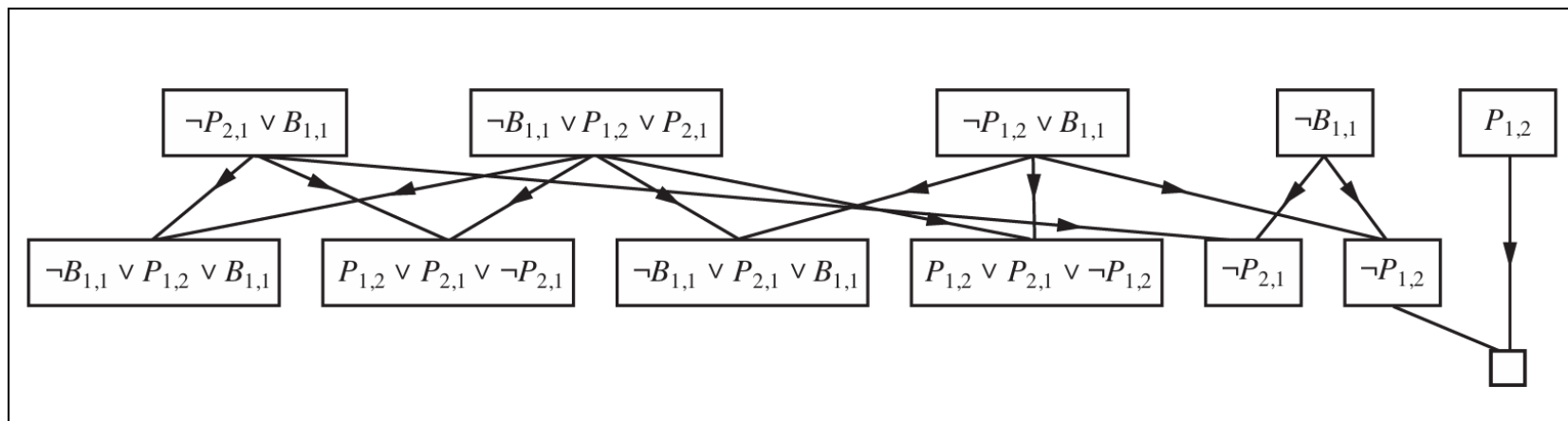
if $new \subseteq clauses$ **then return** *false*

$clauses \leftarrow clauses \cup new$

Resolution – Wumpus World

- Say, the agent is in [1,1], no breeze, so no pits can be in there
- $\alpha = \neg P_{1,2}$

$$KB = R_2 \wedge R_4 = (B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})) \wedge \neg B_{1,1}$$



Thank You

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